

COMPREHENSIVE FITNESS PROGRESS MONITOR ANALYSIS

1. PROJECT OVERVIEW:

The **Comprehensive Fitness Progress Monitor** is designed to provide a holistic view of user health and lifestyle patterns by integrating multiple fitness metrics into a single, interactive dashboard. It consolidates data on calories burned and consumed, sleep quality and duration, step counts, workout types, and average weight to deliver actionable insights. By visualizing correlations between activity, nutrition, and rest, the dashboard empowers users and stakeholders to track progress, identify trends, and make informed decisions for improved wellness outcomes.

2. OBJECTIVE:

- Track overall user health metrics including calories, steps, sleep, and weight.
 - Identify top-performing users based on calories burned and consumed.
 - Analyse workout types to measure their impact on calorie expenditure.
 - Assess sleep quality and duration to understand its effect on fitness outcomes.
 - Compare step counts across different sleep quality categories for behavioural insights.
 - Provide a holistic view of user activity trends to support personalized fitness recommendations.
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3. DATA CLEANING (PREPROCESSING):

The following preprocessing steps were performed to ensure data quality:

- Handled missing values
- Removed duplicate records
- Corrected and changed data types
- Renamed columns for better readability
- Delimit the columns
- Sort and filter the column
- Ensured uniqueness across key identifiers for reliable aggregation.

TOOLS USED:

- Tableau: visualization and dash board creation.
 - Excel → Data cleaning and preprocessing.
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4. KEY PERFORMANCE INDICATORS (KPIs)

These are the important measures in the dashboard:

- **Total Calories Burned:** 3,34,073
- **Total Calories Consumed:** 22,90,924
- **Average Sleep Hours of Users:** 7.05
- **Total Steps Count:** 64,35,338
- **Average Weight of Users:** 79,621
- **Total Fitness Tracker Users:** 40,058

5. VISUALIZATION:

The final dash board of this analysis



6. KEY INSIGHTS (SMART + ANALYTICS APPROACH):

The following KPIs were identified through the analysis.

6.1 DESCRIPTIVE ANALYSIS:

- Users burned 3.34 lakh calories compared to 22.9 lakh consumed, showing a large intake–expenditure gap.
- Average sleep duration is 7.05 hours, with most users in “Good” or “Excellent” categories, but notable hours in “Fair” and “Poor.”
- Running leads calorie burn (95K), followed by weightlifting (70K) and cycling (55K), while yoga and rest contribute minimally.
- Total step count is 64.35 lakh, with higher counts linked to better sleep quality, reinforcing rest–activity correlation.
- Top performers: USR-9021 and USR-9005 lead in calories burned (~10K), while USR-9030 and USR-9049 top calorie intake (~45K).
- Overall engagement is strong with 40,058 fitness trackers and average user weight at 79.6 kg, but lifestyle balance remains a key challenge.

6.2 DIAGNOSTIC ANALYSIS

- The calorie gap exists because users consume far more than they burn, showing insufficient workout intensity or frequency.
- Poor sleep quality reduces energy and motivation, directly lowering workout performance and daily activity levels.
- Running dominates calorie burn due to its accessibility, while low engagement in yoga/HIIT reflects limited workout variety.
- Users with excellent sleep achieve higher step counts, proving rest quality strongly influences physical activity.

- High burners and high consumers highlight diverse lifestyle habits — disciplined vs. indulgent user segments.
- Despite strong tracker adoption, average weight remains high, showing tracking alone doesn't drive results without consistent action.

6.3 PREDICTIVE ANALYSIS

- Average user weight may rise by 5–8% in 6 months if calorie intake continues to exceed burn rates.
- Poor sleepers could reduce activity levels by 10–15%, further lowering calorie expenditure.
- Running will remain dominant, but lack of workout variety may cause performance plateaus within 3–4 months.
- Step count differences will widen, with excellent sleepers maintaining a 20% higher step count than poor sleepers.
- High consumers may increase calorie intake by 5–7%, diverging further from disciplined high burners.
- Despite tracker adoption, only ~30% of users are likely to show measurable health improvements without lifestyle changes.

6.4. PRESCRIPTIVE ANALYSIS

- Reduce Calorie Gap – Implement diet tracking and personalized meal plans to cut average calorie intake by 10% in 3 months.
- Improve Sleep Quality – Introduce sleep hygiene programs and reminders to raise “Excellent” sleep users by 15% within 90 days.
- Diversify Workouts – Promote HIIT and yoga sessions to boost their calorie burn contribution by 20% in the next quarter.
- Increase Step Count – Launch gamified challenges to raise average daily steps by 1,000 per user in 2 months.

- Target High Consumers – Provide nutrition coaching to reduce top users' calorie intake by 5% in 6 weeks.
 - Enhance Tracker Engagement – Add motivational alerts and progress badges to increase active tracker usage by 25% in 6 months.
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7. CONCLUSION:

The analysis rising health risks as calorie intake continues to exceed burn rates, leading to weight gain in the coming months. Poor sleep quality will further reduce activity, while reliance on running may cause workout plateaus. Step count gaps will widen between excellent and poor sleepers, and despite strong tracker adoption, only a small portion of users will achieve measurable improvements without lifestyle changes